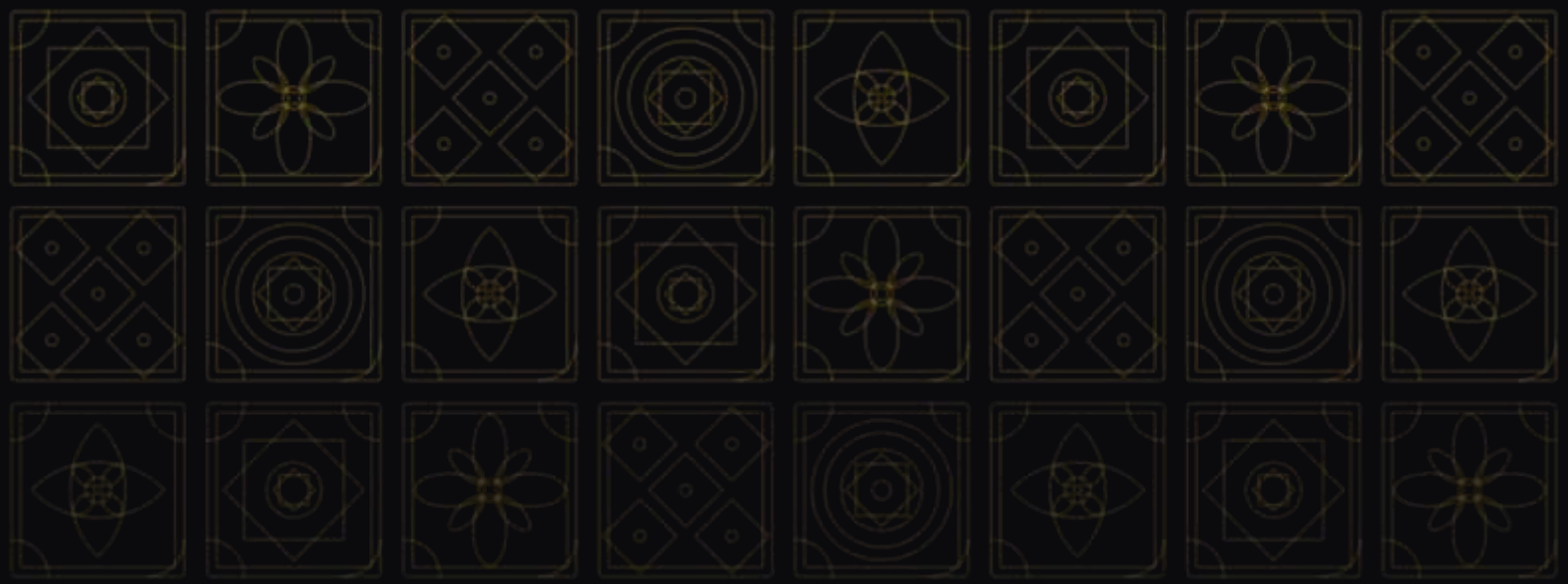




TRAINING PROGRAMME

Machine Learning & Deep Learning

From linear models to Transformers, with the mathematics behind them

75 HOURS + 9 H OPTIONAL

FOR Developers, analysts and engineers who want to build machine learning models



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Overview

This programme teaches participants to build, evaluate and explain machine learning models, and the mathematics that makes them work. They cover supervised learning, from linear and logistic regression to trees, ensembles and support vector machines, with sound evaluation, tuning, pipelines and experiment tracking; unsupervised learning; and deep learning, from a network built from scratch to PyTorch, CNNs, attention and the Transformer. A responsible AI phase covers explainability, fairness and governance, an optional reinforcement learning module adds 9 h, and the programme closes with an end-to-end machine learning project presented by each team.

At a glance

#	PHASE	FOCUS	HOURS
01	Supervised Learning	Regression, classification, trees, ensembles, SVMs, evaluation, pipelines	24 h
02	Unsupervised Learning	Clustering, mixture models, PCA, manifold learning, anomalies, recommenders	12 h
03	Deep Learning	Neural networks from scratch, backpropagation, PyTorch, CNNs, attention and Transformers	15 h
04	Responsible AI	Explainability, fairness, the EU AI Act, model cards	6 h
05	Reinforcement Learning	OPTIONAL MDPs, dynamic programming, Q-learning, DQN, policy gradients	+9 h
06	Final Project	An end-to-end machine learning project, presented to the group	9 h
	Buffer	Catch-up, revision, extra lab time	9 h
	Total	Not included: optional phase 05, + 9 h	75 h

Key facts

AUDIENCE	Developers, analysts and engineers who want to build machine learning models
PREREQUISITES	Python, NumPy and pandas (course: Python for Data Science), or equivalent; secondary-school algebra. The mathematics needed is taught in the course
DURATION	75 h: 66 h of chapters + 9 h buffer; optional reinforcement learning module: +9 h
FORMAT	Lectures that derive each method, followed by labs in Jupyter; the mathematics is taught where it is used
LAB ENVIRONMENT	One laptop per participant (16 GB RAM, Python 3, JupyterLab, scikit-learn, PyTorch, MLflow); GPU notebooks on Google Colab or a GPU server provided by the organiser for deep learning; no paid cloud account
ASSESSMENT	A lab validated at the end of each phase and a final project presented to the group; certificate of completion for participants who complete both
OUTCOME	Participants can frame a machine learning problem, build and compare classical and deep learning models, evaluate them without leakage, explain their predictions and track the experiments

PHASE 01

24 H

Supervised Learning

Participants learn how a model generalises and how to measure it honestly, then build the main supervised methods from their mathematics: linear and logistic regression trained by gradient descent, decision trees and ensembles, and support vector machines. They finish with cross-validation, hyperparameter search, feature pipelines and experiment tracking with MLflow. Validated lab: a tuned scikit-learn pipeline compared with two baselines on a held-out test set, with every run tracked in MLflow.

#	CHAPTER	CONTENT	FORMAT	HOURS
1.1	Framing & Linear Regression	Regression vs classification, generalisation, data splits, data leakage; vectors, matrices, normal equation	Lecture + Lab	3 h
1.2	Gradient Descent & Regularisation	Derivatives, chain rule, gradient descent, ridge and lasso	Lecture + Lab	3 h
1.3	Logistic Regression & Metrics	Sigmoid, maximum likelihood, cross-entropy, precision, recall, ROC	Lecture + Lab	3 h
1.4	Decision Trees	Entropy, information gain, Gini impurity, CART	Lecture + Lab	3 h
1.5	Ensembles	Bagging, random forests, boosting (XGBoost, LightGBM)	Lecture + Lab	3 h
1.6	Support Vector Machines	Margins, soft margin, kernels, SVR	Lecture + Lab	3 h
1.7	Evaluation & Tuning	Cross-validation, learning curves, grid, random and Bayesian search	Lecture + Lab	3 h
1.8	Feature Engineering & Pipelines	Encoding, scaling, scikit-learn pipelines, MLflow tracking	Lecture + Lab	3 h

MATHS COVERED IN THIS PHASE

Linear algebra (vectors, matrices, normal equation) · Calculus (derivatives, chain rule, gradient) · Statistics (maximum likelihood, cross-entropy, bias and variance) · Information theory (entropy, information gain) · Optimisation (convexity, gradient descent, L1 and L2 penalties, constrained optimisation for the SVM margin, kernels)

PHASE 02

12 H

Unsupervised Learning

Participants find structure in data without labels: hierarchical, density-based and centroid clustering, Gaussian mixtures trained with expectation-maximisation, dimensionality reduction with PCA, t-SNE and UMAP, anomaly detection, association rules and collaborative filtering. Validated lab: a customer segmentation with K-Means and a Gaussian mixture, visualised after PCA, with the choice of the number of clusters justified.

#	CHAPTER	CONTENT	FORMAT	HOURS
2.1	Unsupervised Framing	Clustering, density estimation, dimensionality reduction	Lecture + Lab	1.5 h
2.2	Hierarchical Clustering & DBSCAN	Linkage criteria, density-based clustering	Lecture + Lab	1.5 h
2.3	K-Means & Gaussian Mixtures	Objective function, Lloyd's algorithm, expectation-maximisation	Lecture + Lab	3 h
2.4	PCA	Covariance, eigendecomposition, explained variance	Lecture + Lab	3 h
2.5	Manifolds & Anomalies	t-SNE, UMAP, isolation forest	Lecture + Lab	1.5 h
2.6	Association Rules & Recommenders	Apriori, support, confidence and lift, collaborative filtering	Lecture + Lab	1.5 h

MATHS COVERED IN THIS PHASE

Probability (multivariate Gaussian, expectation-maximisation) · Linear algebra (covariance matrix, eigendecomposition) · Distances and similarity measures

PHASE 03

15 H

Deep Learning

Participants build a neural network in NumPy to see every step, derive backpropagation, then train models with PyTorch: convolutional networks and transfer learning for images, recurrent networks for sequences, attention and the Transformer architecture. Validated lab: an image classifier fine-tuned from a pretrained CNN in PyTorch, with its training curves and checkpoints.

#	CHAPTER	CONTENT	FORMAT	HOURS
3.1	Perceptron & MLP from Scratch	NumPy implementation of a multilayer perceptron	Lecture + Lab	3 h
3.2	Backpropagation	Computational graphs, vanishing gradients	Lecture + Lab	3 h
3.3	Training with PyTorch	Tensors, autograd, training loops, optimisers (SGD, Adam), regularisation, checkpoints	Lecture + Lab	3 h
3.4	CNNs	Convolution, pooling, architectures, transfer learning	Lecture + Lab	3 h
3.5	Sequences & Attention	RNN, LSTM and GRU, attention, the Transformer architecture	Lecture + Lab	3 h

MATHS COVERED IN THIS PHASE

Calculus (chain rule on computational graphs) · Linear algebra (matrix products, convolution) · Optimisation (stochastic gradient descent, Adam) · Probability (softmax, cross-entropy)

PHASE 04

6 H

Responsible AI

Participants explain model predictions globally and locally, measure and mitigate bias, and document models for governance under the EU AI Act and data protection rules. Validated lab: a SHAP explanation and a fairness report for the phase 1 model, summarised in a model card.

#	CHAPTER	CONTENT	FORMAT	HOURS
4.1	Explainability	SHAP, LIME, feature importance	Lecture + Lab	3 h
4.2	Bias, Fairness & Governance	Fairness metrics, the EU AI Act and data protection, model cards	Lecture + Workshop	3 h

PHASE 05 · OPTIONAL

+9 H

Reinforcement Learning

This optional module adds 9 h for groups who want it. Participants model decision problems as Markov decision processes, solve them with dynamic programming, then learn from experience with Q-learning and SARSA, deep Q-networks and policy-gradient methods. Validated lab: an agent trained with Q-learning and then with a DQN on a Gymnasium environment, with the learning curves compared.

#	CHAPTER	CONTENT	FORMAT	HOURS
5.1	RL & MDPs	Agent, environment, rewards, policies, Bellman equations	Lecture + Lab	1.5 h
5.2	Dynamic Programming	Value iteration and policy iteration	Lecture + Lab	1.5 h
5.3	Q-Learning & SARSA	Temporal-difference learning	Lecture + Lab	1.5 h
5.4	Deep Q-Networks	Experience replay, target networks	Lecture + Lab	1.5 h
5.5	Policy Gradients & Actor-Critic	REINFORCE, A2C, PPO overview	Lecture + Lab	3 h

MATHS COVERED IN THIS PHASE

Probability (Markov decision processes, expectations) · Dynamic programming (Bellman equations) · Calculus (policy gradient theorem, log-derivative trick)

PHASE 06

9 H

Final Project

Teams of two or three carry a machine learning problem from framing to explanation on a provided dataset, reusing the phase labs, then present it as they would to a product team.

#	CHAPTER	CONTENT	FORMAT	HOURS
6.1	Framing & Exploration	Frame the problem and the metric, exploratory data analysis, split strategy	Project	3 h
6.2	Models & Experiments	Feature pipeline, comparison of classical models and a deep learning model, all tracked in MLflow	Project	3 h
6.3	Explain & Document	Explainability report, model card, short technical report	Project	1.5 h
6.4	Presentation & Review	Presentation of the results and limits; questions and feedback	Project	1.5 h

FINAL PROJECT

End-to-End Machine Learning Project

A machine learning problem carried from framing and exploration to a feature pipeline, a comparison of classical models with a deep learning model, an explainability report and a model card, with every experiment tracked.

STACK Python, pandas, scikit-learn, XGBoost or LightGBM, PyTorch, MLflow, SHAP, JupyterLab

DELIVERABLE Repository with notebooks and pipeline code, MLflow runs, explainability report, model card, presentation